Venkata Lakshmi Parimala Pasupuleti

AI Engineer · Agentic Document Pipelines · Income Verification · OCR Extraction · Durable Async Workflows · LangGraph · LLM Evaluation · Python · FastAPI · AWS · Production AI

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PROFESSIONAL SUMMARY

AI engineer with two shipped agentic systems and 8 years of production Python experience directly matched to Balerion's technical problems. Owned income qualification pipelines end-to-end for 7 years at Entrata: federal HUD source → 180,000 income-limit domain records powering housing eligibility decisions across 100K+ units - zero data integrity incidents, 100% PHI compliance, 99.5% uptime. The invariant was the same as mortgage origination: a wrong income number is far worse than a missing one. LeaseI is an 8-agent LangGraph stateful system with OCR document extraction, multi-source RAG retrieval, deterministic guardrails on regulated outputs, and HITL escalation - 93.6% precision across 1,240 compliance test cases. AgentProbe is an open-source LLM evaluation framework designed for evaluating multi-step agent outputs against ambiguous criteria: LLM-as-judge scoring, 5-metric quality engine, CI-integrated regression gate - zero regressions shipped across 1,240+ test cases. Async resilience at Entrata mirrors Restate's durable execution model: RabbitMQ event-driven processing with retry, idempotency, and dead-letter guarantees across 7yr production pipeline. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026).

TECHNICAL SKILLS

AI & Agentic Systems: Agentic Workflows, LangGraph, LangChain, Multi-Agent Orchestration, RAG Pipelines, OCR Document Extraction, LLM Evaluation Frameworks, LLM-as-Judge, CI-Integrated Regression Gating, Offline Evals, HITL, Guardrails, Structured Outputs, Prompt Engineering, ChromaDB, Vector Search, Embeddings, Anthropic API (Claude), OpenAI API, Groq, Ollama, Claude Code, MCP, BERT, spaCy, NLP

Engineering & Data Pipelines: Python (8yr), FastAPI, SQL, TypeScript, JavaScript, Docker, Microservices, REST APIs, CI/CD (GitHub Actions), AWS (S3, EC2, RDS, Glue), RabbitMQ (retry, idempotency, dead-letter), Redis, PostgreSQL, MySQL, Oracle, Bash, Linux, Git

Production & Compliance: Durable Async Processing, Income Verification Pipelines, PHI Compliance (Libsodium), Data Integrity, Observability, Performance Engineering, PyTorch, scikit-learn, NumPy, Pandas, PySpark, Feature Engineering, Statistical Modeling, Agile/Scrum

EXPERIENCE

Magizhs Tech Solution

May 2025 - Aug 2025

AI/ML Engineer Intern

- Built agentic document processing pipeline (LangGraph, Python) for mission planning - OCR and structured extraction over multi-source documents; deterministic guardrails before regulated outputs; HITL escalation on low-confidence extraction results.
- CI/CD-integrated eval gate enforcing quality on every release; observability on extraction accuracy, retrieval latency, and agent performance.

George Mason University

Aug 2024 - May 2026

Graduate Teaching Assistant & Research Associate

- Built Python ML pipelines (PyTorch, scikit-learn) on WHO and UK Road Safety datasets - $R^2 = 0.977$; end-to-end from data ingestion through feature engineering and statistical validation across 190+ countries.
- Mentored 600+ students in Python and data science; designed monitoring dashboards improving research comprehension by 40%.

Entrata - PropTech / Affordable Housing SaaS

Sep 2018 - Aug 2024

Senior Software Engineer · Lehi, UT (Remote)

- **Income qualification pipeline ownership - 7 years** - owned HUD income-limit pipeline end-to-end (federal source → 180,000 domain records powering eligibility and income compliance for 100K+ housing units); the invariant: a wrong income number is far worse than a missing one - enforced with 100% PHI compliance (Libsodium-encrypted), zero data integrity incidents, 99.5% uptime SLA.
- **Durable async processing with retry and idempotency** - RabbitMQ event-driven pipeline with retry, idempotency, and dead-letter guarantees across 7yr production workflow (= durable execution model); Redis caching; Docker containerization (50% scalability improvement); AWS S3, EC2, RDS at enterprise scale.
- **AI-driven observability and engineering standards** - AI-powered monitoring reducing triage time 40%; 200+ SQL query optimizations (30% gain); 300+ code reviews; CI/CD via GitHub Actions; mentored 5+ engineers on production AI and backend best practices.

Entrata India - Pune

Jan 2017 - Sep 2018

Associate Software Engineer → Trainee

- Built backend modules (PHP, Python, PostgreSQL); data pipeline for 1.6M HUD/Tax Credit records; AWS S3 migration resolving 90% of legacy storage constraints.

PROJECTS

AgentProbe: LLM Evaluation for Multi-Step Agent Outputs

Evals · Ambiguous Criteria

Python · FastAPI · LangGraph · ChromaDB · CI/CD · Anthropic · OpenAI · Ollama | github.com/venkatalakshmiparimala/AgentProbe

Production LLM evaluation framework purpose-built for evaluating multi-step agent output against ambiguous compliance criteria: LLM-as-judge scoring, 5-metric quality engine (correctness, groundedness, latency, token efficiency, hallucination rate), CI-integrated regression gate, multi-provider Anthropic/OpenAI/Ollama routing. Zero regressions shipped across 1,240+ test cases - evaluation infrastructure built before any prompt reaches production.

Leasel: 8-Agent Agentic Document Processing & RAG System

Agentic · OCR · RAG · HITL

Python · LangGraph (8-agent) · FastAPI · TF-IDF · OCR · SQLite | github.com/venkatalakshmiparimala/Lease_complaince_rag

Production agentic AI system with document extraction pipeline: OCR over scanned lease documents, multi-source RAG retrieval, 8-agent LangGraph stateful workflow with routing, tool use, and state management; deterministic guardrails on every regulated compliance output; HITL escalation when model confidence below threshold; source-linked answers with 100% citation correctness. 93.6% precision, 87.9% F1 across 1,240 compliance test cases.

AI on FHIR: NLP-Driven Medical Records Query System

RAG · NLP · Extraction

Python · Flask · LLMs · FHIR · React · Docker · spaCy · BERT · Embeddings | github.com/venkatalakshmiparimala/AI_on_FHIR

RAG pipeline with structured data extraction over proprietary FHIR records: BERT embeddings + LLM inference translating natural-language queries into precise FHIR API calls - reduced retrieval time ~60%; source-linked outputs with HIPAA-compliant RBAC and full data isolation.

EDUCATION

M.S. Data Analytics Engineering - George Mason University, Virginia — May 2026 · GPA 3.87

B.Tech Computer Science & Engineering - JNTU Anantapur — 2013-2017